

High energy and high power density supercapacitor with 3D Al foam-based thick graphene electrode: Fabrication and simulation



Zhoufei Yang^{a,1}, Jiarui Tian^{a,1}, Zhenzhen Ye^a, Ying Jin^{a,b}, Chaojie Cui^{a,**}, Qing Xie^a, Jin Wang^a, Gang Zhang^b, Zhuoya Dong^a, Yonghua Miao^b, Xiang Yu^a, Weizhong Qian^{a,*}, Fei Wei^a

^a Department of Chemical Engineering, Tsinghua University, Beijing, 100084, PR China

^b Jiangsu Zhongtian Technology Co., Ltd., Nantong, 226463, Jiangsu, PR China

ARTICLE INFO

Keywords:

Supercapacitor
Thick electrode
Al foam
Graphene
Ion polarization

ABSTRACT

Thick electrode was promising to increase the energy density in device (battery or supercapacitor) scale, but always suffered the ion polarization upon high rate discharge. We reported the use of 3D Al foam (current collector) to host graphene (with high surface area but very low packing density) to fabricate 450 μm thick electrode with mass loading of 10–15 mg cm^{-2} . The space confinement effect of Al wires on graphene at any regions offered sufficient ordered diffusion channel of liquids, alleviating the ion polarization significantly, compared to the same thick graphene sheet pasted on 2D Al foil by simulation with COMSOL software. The 100 F pouch supercapacitor with ionic liquids at 4 V exhibited low contact resistance, high specific capacitance in a wide range of current densities, and simultaneously the volumetric energy density of 18 Wh L^{-1} and the power density of 5 kW kg^{-1} , far better than those of current commercial device.

1. Introduction

Supercapacitor (SC) was a typical electrochemical energy storage device with high power density, but suffered from relatively low energy density, which limited its application fields [1–3]. Increasing the energy density called for the electrode with high capacitance and stable operability when working at high voltage, as well as the high mass or volume ratio of electrode inside the device (e.g. the use of high-density electrode active materials or the use of thick electrode) [4–6]. Graphene, an sp^2 hybridized carbon, exhibited excellent electrical conductivity, high chemical stability and a specific capacitance of 160–200 F g^{-1} at 4 V [1–3,7–9]. But the low packing density resulted in the low mass ratio of it in SC [10]. Densification strategy of graphene, including the solvent evaporation, direct compression or the fine-tuning of the interlayer distance followed by the compression, was the necessary step to enhance the applications, but was far from mature to this date [11,12]. Most of SCs based on thick graphene electrodes suffered the poor ion diffusion property [13] and the associated poor rate performance. Therefore, it remained a great challenge for the use of graphene electrode toward

commercialization.

Herein, we proposed a strategy to provide high density graphene sheet but with sufficient and ordered diffusion channels of electrolyte ions. In detail, three-dimensional (3D) Al foam (current collector) with a thickness of 1 mm was used to host graphene slurry followed by the roll compression. The as-made 350 μm thick graphene electrode, with a mass loading of graphene of 10–15 mg cm^{-2} , exhibited smooth expansion property to 450–480 μm as the intaking electrolyte. The as-made pouches of 10–500 F exhibited far higher volumetric energy density, due to the high volume contribution of the thick electrode in the device, as compared to commercial SCs with 100–130 μm thick electrode pasted on Al foil. As an example, the 100 F pouch exhibited energy density of 18 Wh L^{-1} and power density of 5 kW kg^{-1} in ionic liquid at 4 V. The model of 3D electrode structure was built and simulated with COMSOL software and compared with the thick graphene electrode pasted on 2D Al foil, suggesting that the provision of sufficient channels of electrolyte ions by the ordered 3D foam structure prevented graphene from forming large layers, allowed the full access of ions to electrode material at high current densities, alleviated the ion polarization across the thick electrode

* Corresponding author.

** Corresponding author.

E-mail addresses: cuij06@163.com (C. Cui), qianwz@tsinghua.edu.cn (W. Qian).

¹ These authors contributed equally to this work.

significantly, and increased the capacitance effectively. Meanwhile, the Al foam-enhanced graphene-based SCs with high volumetric energy density was easily fabricated, very promising for the space-constrained applications.

2. Experimental section

2.1. Preparation of graphene and electrode

Al foam: The 3D Al foam was prepared with a physical evaporation method reported elsewhere [14]. The necessary steps included the deposition of Al layer on the substrate of porous polyurethane membrane with 1 mm thick, and then totally removed the polymer by burning with air at 350–500 °C (5 °C/min) for 6 h. The as-obtained Al foam was cooled to ambient with Ar gas and kept in a condition free of oxygen and water. The 3D network, consisted of pure Al wires of around 0.15 mm in diameter, had the porosity of 96%. The average pore size was about 0.5 mm, the thickness was 1 mm and the areal density was 140 g m⁻².

Graphene: Graphene was synthesized by the chemical vapor deposition method reported elsewhere¹⁹. With the catalyst of metal-supported Layered Double Hydroxide (LDH), graphene-carbon nanotube hybrids (GNHs) in the presence of metal or pure graphene in the absence of metal were prepared. The detailed condition included: space velocity of methane: 100 ml g cat⁻¹ h⁻¹, gas feedstock (by volume ratio): methane:H₂:Ar = 1:2:1. Temperature: 950 °C. Pressure: ambient pressure. Reactor: quartz tube with an inner diameter of 30 mm. Reaction time: 2 h. Graphene product was obtained after removing the substrate by acid solution, followed by washing with deionized water for 5 times, and drying by freeze-drying at -25 °C for 24 h. The graphene was further treated by a high-temperature metal impurity removal process at 10⁻² Pa for 12 h.

Electrode: The graphene slurry was made by a grinding process during which about 10 g graphene, 0.77 g Polyvinyl Pyrrolidone (PVP) with or without PTFE were mixed in 700 g deionized water, followed by grinding at 2000 rpm for 4 h. A typical photograph of the graphene slurry could be found in Fig. S1A (SI-1). To obtain the electrodes, the slurry was uniformly coated on Al foil or filled in the pores of the 3D Al foam under vacuum condition, dried and compressed with a roll-compression under 10 MPa into 450 μm. The areal mass loading of graphene in Al foam electrode sheets was approximately 10–15 mg cm⁻².

2.2. Fabrication of coin-type and pouch SCs

Fabrication of coin-type SCs: The obtained electrode sheets were cut to be round disks with a diameter of 13 mm. The disks were dried under vacuum at 150 °C for 12 h and then transferred into a glove box. Two disks with the same mass loading were assembled with TF4030 membrane (NKK corporation) and filled with excess EMIMBF₄ (Shanghai chengjie Chemical) electrolyte in 2025 type coin shell.

Fabrication of pouch SCs: The obtained electrode sheets were cut to be a rectangle with the sizes of 3 cm*5 cm (for the pouch of 100 F and lower) or 7 cm*10 cm (for the pouch of 100 F–500 F). After assembling with TF4030 membrane and pole ear welding, 16 pieces of rectangular electrode sheets of about 450 μm thick graphene-based Al foam electrode were dried under vacuum at 150 °C for 12 h and then transferred into glove box, assembled with aluminum-plastic package, filled in EMIMBF₄ electrolyte under vacuum and sealed. The weight of the EMIMBF₄ electrolyte added was around 5.8 times than that of the total mass loading of active materials in pouch.

2.3. Characterizations

The morphology and microstructure of the graphene and Al foam were observed by field-emission scanning electron microscopy (JSM-7401) and transmission electron microscope (JEM-2010). The wet electrode was observed by environmental scanning electron microscopy (FEI QUANNTA 200 FEG). Nitrogen adsorption/desorption isotherms were

performed using a surface area and porosity analyzer (QUADRASORB SI) at 77 K. The cycling charge/discharge test and electrochemical impedance spectroscopy (EIS) are measured on an electrochemical workstation (VMP 300, Biologic). Potentiostatic EIS is performed over a frequency range of 100 kHz to 0.01 Hz with an applied amplitude of 5 mV.

2.4. Calculations

The specific capacitance of active materials graphene C_{gr} (unit: F/g) could be calculated as follows:

$$C_{gr} = \frac{8I \int U dt}{U_{max}^2} \quad (1)$$

where I (unit: A g⁻¹) was the applied constant current density of EDLC based on all the material in EDLC, t (unit: s) was the time in the charge/discharge curve. U (unit: V) is the instant voltage in the charge/discharge curve. U_{max} (unit: V) was the maximum voltage of devices ($U_{max} = 4$ V in this work).

Particularly, When the curve of galvanostatic discharge shown linearity, equation (1) could be simplified as follows:

$$C_{gr} = \frac{4It}{U_{max}} \quad (2)$$

The maximum power density of the device P_V (unit: kW L⁻¹) or P_M (unit: kW kg⁻¹) was calculated by:

$$P_V = \frac{U_{max}^2}{4 \times \frac{\Delta U}{\Delta I} \times V_{device}} \quad (3)$$

$$P_M = \frac{U_{max}^2}{4 \times \frac{\Delta U}{\Delta I} \times M_{device}} \quad (4)$$

where ΔU (unit: V) and ΔI (unit: A) were the absolute value of instantaneous voltage/current difference at initial discharge of 30 ms, V_{device} (unit: cm³) is the volume of the device. M_{device} (unit: g) is the mass of the device.

3. Results and discussion

3.1. The design and characterization of the Al foam-based SC electrode

The graphene product was dominated by 1–3 layers (Fig. 1A). It had a specific surface area of 1443 m² g⁻¹ with obvious mesopores of 3–30 nm (Fig. 1B and C). Hollow cages in the sample were due to the removal of the MgO substrate totally, which made it relatively easy to grind into small monodispersed graphene sheets (Fig. 1D). Graphene slurry was relatively sticky due to the high volume ratio of graphene (Fig. 1E), which was the key for filling into Al foam without leaking too much. Fig. 1F showed the uniform 3D Al foam with a length of 20 cm, which was nearly transparent since its thickness was only 1 mm and porosity was 96%. The strain-stress curve showed that the highest strength could be 0.57 MPa at the strain of 4.8% (Fig. S1B, SI-1). The cracking of the foam occurred at the strain of 5%–6%, with a slow decrease of stress and the obvious elongation. Apparently, the Al foam exhibited high strength and structure flexibility in the fabrication of thick electrode as hosting the graphene slurry and upon compression. Pristine Al foam consisted of many interconnected Al wires which diameter was very uniform (Fig. 1G). After compressed to 0.35 mm thick, all interconnected Al wires remained unbroken and the diameter of each Al wire was unchanged (Fig. 1H), which was the key to keep the strength of the graphene electrode. The electrode of graphene-Al foam was totally black at the appearance, since the outer surface of Al wires was covered by graphene. The sketch of Al wires could be visible after slightly scratching out of the outer layer of graphene. Energy Dispersive X-Ray Spectroscopy (EDS)

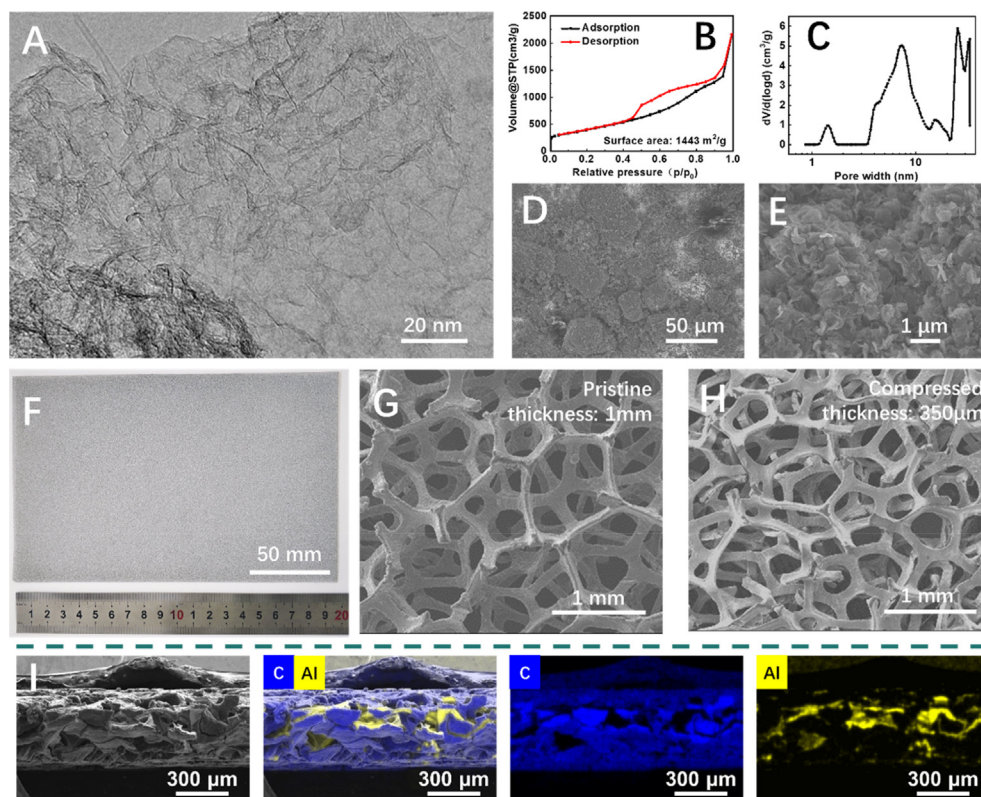


Fig. 1. Characterization of graphene, Al foam and graphene-Al foam electrode. (A) TEM image of graphene used in the present work. (B) Nitrogen isothermal adsorption curve of graphene. (C) The pore size distribution of graphene. (D) SEM image of graphene after grinding. (E) SEM image of graphene slurry. (F) The photo image of Al foam. (G) SEM image of pristine Al foam. (H) SEM image of Al foam after compression. (I) EDS mapping images of the cross-section of the graphene-Al foam electrode.

mapping images of the cross-section of the dry composite electrode confirmed the presence of C and Al elements (Fig. 1I). The Al foam structure was uniformly filled with graphene, showing the high mass loading and dense packing of active materials, which enabled the high

energy density. To deeply investigate the element distribution of different positions, the single point EDS test was also performed (Fig. S2, SI-2). The overlap of C and Al element at position 1 and position 3 was due to the complete wrapping of Al wires by the small-sized graphene,

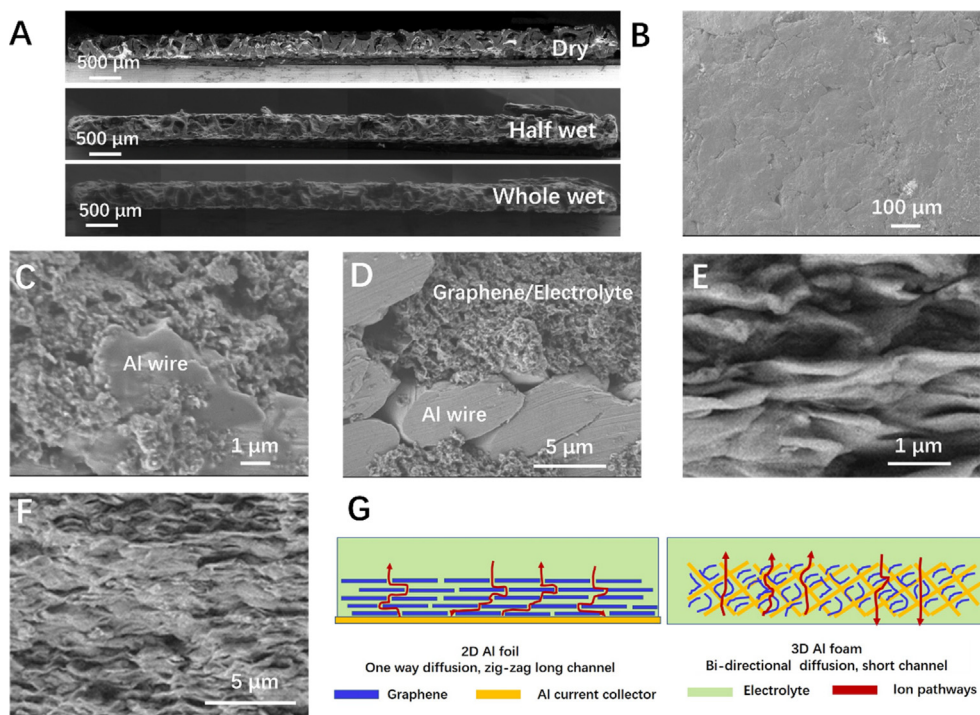


Fig. 2. Characterization of swelling behavior of graphene-Al foam electrode. (A) SEM images of graphene-Al foam electrode in a dry state, half wet state and wet state. (B) SEM image of the outer surface of a wet electrode. (C and D) SEM images of a cross-section of the wet graphene-Al foam electrode. (E and F) SEM image of compressed graphene-Al foil electrode. (G) Schematic diagram of ion diffusion behavior in the electrode assisted by Al foil or Al foam.

which showed the good contact behavior between the graphene and Al surface. Note that only graphene with very small and uniform size allowed the sufficient filling of graphene inside and the excellent contact with the sketch of the current collector.

3.2. The swelling behavior of graphene electrode

The environmental scanning electron microscope was used to observe the structure change of the electrode as filling electrolyte gradually. Fig. 2A showed a very long dry electrode with obvious holes, due to the volume shrinkage of graphene. Its surface and thickness across the long electrode remained very stable as filling electrolyte, whether in half wet state or whole wet state. It suggested that the presence of numerous Al wires exhibited the high mechanical strength inside the graphene electrode, which uniformly constrained and distributed the swelling force of graphene, ensuring the steady and uniform volume expansion of graphene. Quantitatively, the thickness of the special electrode was 351.2 μm at the beginning dry state and eventually increased up to 488.5 μm after 720 min, as filling 1-ethyl-3-methylimidazolium tetrafluoroborate (EMIMBF_4) electrolyte 5.8 times the mass of graphene. The wet electrode had a very smooth surface (Fig. 2B). Cross-section of the wet electrode showed the tight contact of Al wires with graphene (Fig. 2C and D). Compression of the electrode made some Al wires contact together, implying high compression force also made graphene in Al wire cages contact together.

In sharp contrast, if without Al wires, the tightly compressed graphene-Al foil sheet tended to form large and parallel layers (Fig. 2E and F), which was the nature of 2D materials. In theory, the larger the size of graphene was, the easier it was to stack, and the more parallel layers were formed. Such a structure in the microscopic scale resulted in the abrupt swelling in the local region as intaking electrolyte (Fig. S3A, SI-3), which was undesirable in fabricating SC. This comparison, in turn, suggested the uniform thickening of the electrode of graphene-Al foam was due to the presence of short and ordered diffusion pathways, allowing all the pores of graphene accessible to the ions of electrolyte quickly (Fig. 2A and Fig. S3B, SI-3). The parallel layers were perpendicular to the diffusion direction of ions upon charge and discharge, resulting in the formation of zig-zag ion pathway channels. The diffusion

pathways of electrolyte ions in different electrodes were summarized in Fig. 2G.

3.3. Electrochemical Performance of SCs

Note that, the flat and uniform thickness of the wet electrode of graphene in Al foam was the key to fabricate pouch smoothly (Fig. 3A and B) since a device consisted of many stacked electrodes. Besides, controlling the expansion of graphene inside Al foam with a low degree facilitated the manufacture of the device with high volume ratio of active materials, compared to that of the graphene-Al foil sheet, as discussed below.

As follows, considering the pure graphene powders were hard to make a thick coating on Al foil to fabricate a pouch, we compared the contact resistance of three kinds of graphene electrodes with coin-cell type SCs. Electrochemical impedance spectroscopy (EIS) was obtained (Fig. 3C). The line of the SC with graphene-Al foam was very straight in the low-frequency region in the Nyquist plot, indicating the tight contact between Al foam and graphene without the necessity to use Polytetrafluoroethylene (PTFE, binder). The use of PTFE enhanced the interaction of graphene with Al foam, but the capacitive response of SC with graphene-Al foil-PTFE deviated the most from the ideal EDLC. In addition, the semicircle of three SCs suggested that the larger resistance for those with PTFE than without PTFE. Quantitatively, the key electrochemical impedance parameters including the electrical resistance of the bulk phase of the electrolyte (R_s), transfer resistance of electric charge (R_{ct}), and transfer resistance (M) were fitted using a the transmission line model of the porous electrode, which was invented by de Levie [15] and developed into a modified Warburg element [16]. The fitting results of EIS curves was shown in Fig. S4 and Table S1 (SI-4). R_s values of three coin cells were nearly the same, since the electrode materials and electrolyte were the same. However, R_{ct} values of the graphene-Al foil-PTFE and of graphene-Al foam-PTFE were 4 and 2.3 times that of the graphene-Al foam, respectively. Apparently, the addition of PTFE exhibited the negative effects on the electron transfer within graphene, as well as blocking the pores of graphene or decreasing the pathways of ions. In addition, the electrolyte phase resistance of Al foam-assisted electrode is much lower than that of Al foil-assisted electrode,

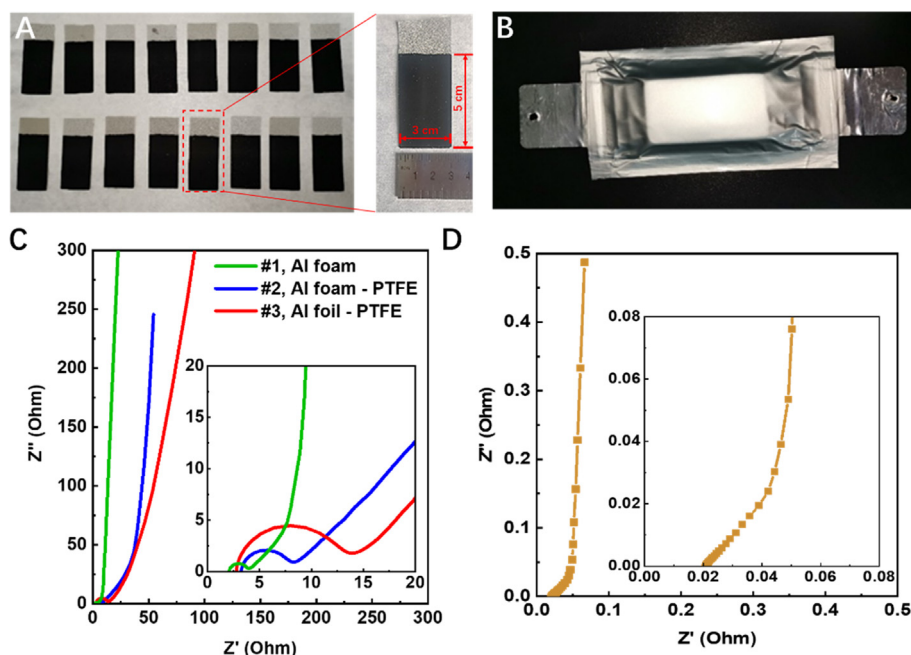


Fig. 3. EIS characterization of graphene-Al foam electrode and graphene-Al foil electrode. (A) Photo images of graphene-3D Al foam electrode sheets of 3 cm*5 cm. (B) Photo of 100F SC pouch, made of the electrode in (A). (C) EIS curves of coin cell with three kinds of graphene electrodes made of Al foam without PTFE, Al foam with PTFE and Al foil with PTFE. (D) EIS curve of the 100F pouch in (B).

validating the 3D Al foam-based electrode provided more ion diffusion channels with lower tortuosity (Fig. 2G). As a result, as integrating 16 pieces of graphene-Al foam electrodes together, the 100 F pouch (Fig. 3A and B) exhibited a much smaller resistance as shown in EIS spectra (Fig. 3D). The R_s value was only $0.0196\ \Omega$, since the stacking of multi-layer electrodes was effective to decrease the contact resistance. R_{ct} value approached to zero, verifying the parallel effect of multi-layer electrodes. The straight tail in the low-frequency region indicated the nearly ideal double electric layer capacitor response of the pouch.

As a result, the 100 F pouch was easily charged to 4 V with the galvanostatic charge and discharge mode (named mode A) (Fig. 4A). The voltage drop at the beginning of the discharge was very small, further validating the low resistance of the pouch. The nearly linear response of charge and discharge voltage curve of the pouch indicated that the ion diffusion inside the graphene in pouch behaved very well. In addition, the pouch was charged and discharged with a mode similar to the standard of capacitor for electrical vehicles [17]. The device was firstly charged with a constant current to 4 V, and then continue to charge as maintaining 4 V constant for a certain time, and finally, the device was discharged with a constant current (named mode B) (Fig. 4B) [18]. The specific capacitance calculated (based on the graphene material) was 132, 130.5, 129, 124 $F\ g^{-1}$ under the current density of 0.2, 0.5, 1.0, 2.0 $A\ g^{-1}$, respectively (Fig. 4C). The values were all larger than those calculated from the mode A. The constant-voltage charge at 4 V was effective to drive ions access to the micropores suffering the barrier of ions in high current density. Energy density and power density were calculated based on the charge mode B, recommended by the Chinese industry standard (Fig. S5, SI-5). Quantitatively, the volumetric energy density of the pouch ranged from 10 to 21 $Wh\ L^{-1}$ with increasing the capacity of the pouch (by increasing the size and mass loading of each electrode) from 5 F to 500 F (Fig. 4D).

The changing trend was quite different from that of activated carbon (AC) based SC with the same capacity, where the small device gave a much lower energy density. The reason was the 2D structure of the

electrode employed by the commercial AC-based SC, which made from the active material coated on two sides of Al foil (100–130 μm) and, had the lower ratio of active materials in the device, due to the thinner electrode. In comparison, thick electrode used in the present work allowed the high ratio of active materials with the high specific capacitance of graphene. Quantitatively, the energy density, calculated based on the whole 100 F pouch devices, was $18.34\ Wh\ L^{-1}$ and $7.65\ Wh\ kg^{-1}$, respectively. The maximum power density (calculated by Equations (3) and (4), detailed calculation steps could be found in Table S2-1, SI-7) was in the range of $11.62\text{--}13.59\ kW\ L^{-1}$ and $4.85\text{--}5.66\ kW\ kg^{-1}$ (Fig. 4E and Fig. S6B, SI-6), based on the volume or mass of the 100 F pouch. Different from the trend in coin cell, power density first decreased, then increased with current density in pouch (Fig. 4E). The large amount of Joule heat produced by a large number of electrode materials probably resulted in the temperature increase in pouch under high current density. It would make the viscosity of EMIMBF₄ decrease and the mobility of ions increase. In this case, the value of power density would be a trade-off effect of the decreased diffusion resistance of EMIMBF₄ under high temperature and the serious ion polarization under high current density. Quantitatively, the volumetric energy density of the 100 F pouch was nearly 2–3 times than that of the commercial AC-based SCs within the same capacity (100 F, organic electrolyte, 2.7 V, Al foil as current collector, Fig. 4D) [19–23]. The gravimetric energy density of our pouch was 1.5–2 times that of the latter, and even comparable to that of commercial AC-based SC of 3000 F (Fig. S6A, SI-6). If compared with commercial devices with similar capacity (50–100F), the present device exhibited nearly the same gravimetric power density, but exhibited a volumetric power density far higher than the commercial device (Table S2-2, SI-7). The comparison validated the 3D Al foam-enhanced device exhibited high energy density and high power density simultaneously. Since the capacity of SC always increased with its size, as minimizing the impact of any components not contributing to the capacitance, it could be expected that our larger pouch might have much excellent performance. Cycling stability was performed for the 100 F pouch. The capacitance retention

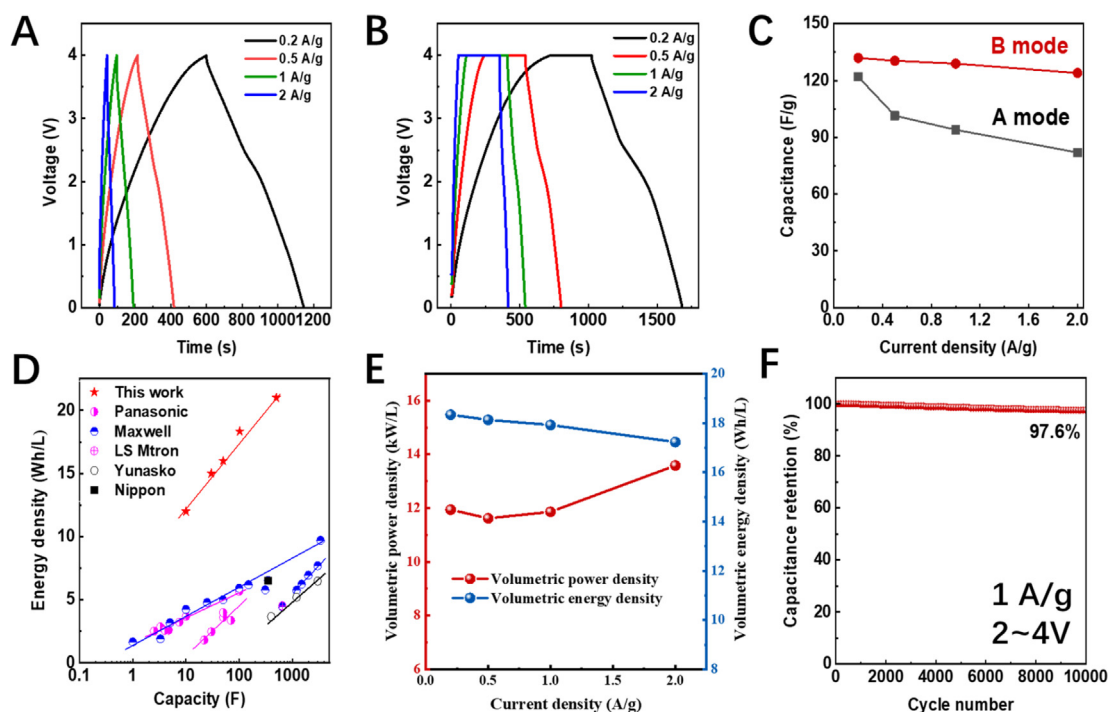


Fig. 4. Electrochemical Performance of supercapacitor devices. (A) Galvanostatic charge-discharge curves of the 100 F pouch. (B) The new constant-voltage charge-discharge curves of the pouch. (C) The specific capacitance of graphene-based on different charge modes in (A) and (B). (D) Comparison of the volumetric energy density of the pouch with various commercial SC devices (Panasonic [22], Maxwell [20], LS Mtron [19], Yunasko [23], Nippon [21].) with active carbon electrode, operating at 2.7 V. The gravimetric energy density comparison plot could be found in Fig. S6A (SI-6). (E) Relationship between the volumetric energy density and volumetric power density of 100 F pouch at different current density. (F) Capacitance retention of the SC pouch for 10,000 cycle tests.

was 97.6% after 10,000 cycles at 1 A/g under 2–4 V (pouch, Fig. 4F). It suggested the Al foam-based device was very promising for commercial use. Further investigation was underway. On the other hand, if one decreased the loading of graphene on Al foam gradually, the as-fabricated electrode with the set thickness of 300 μm exhibited a gradually increased rate performance, as testing a series of 4-sheet based pouches (Fig. S7, SI-8). The power density of 4-sheet pouch is insensitive to the change of current density from 2 A g^{-1} to 5 A g^{-1} , validating again the effectiveness of 3D conductive network of Al foam.

3.4. Electrochemical simulation of the Al foil and Al foam-based devices

Note that, we were unable to experimentally compare the SC with the same thick graphene sheet pasted on Al foil, which was too difficult to fabricate yet. However, the use of a simulation tool allowed the understanding of the diffusion behavior of ions and the associated performance

with different electrode-based SCs. COMSOL software was effective to provide the ability of structure simulation integrated with the porous electrode mode. A detailed model description and simulation could be found in SI-9. We set the simulation parameters according to the property of the electrode and electrolyte used in the present work, and the size of the device (Table S3, SI-9). The electrical conductivity of graphene [24] was set as 3000 S m^{-1} , while it was only 1.57 S m^{-1} for the electrolyte of EMIMBF₄ [25], apparently an ion diffusion-limited process. Since the small-sized graphene was tightly compressed to form paralleled layers, the Bruggeman coefficient was set to be 3, representing the zig-zag ion channel structure inside compressed graphene sheet. No binder was used to avoid the complicated pore blocking effect, which gave an optimistic estimation on the performance of the graphene sheet pasted on Al foil (where a binder is necessary), but gave a close estimation on that of Al foam assisted graphene electrode. The mode considered the structure of electrode, where graphene-Al foil sheet tended to form paralleled layer

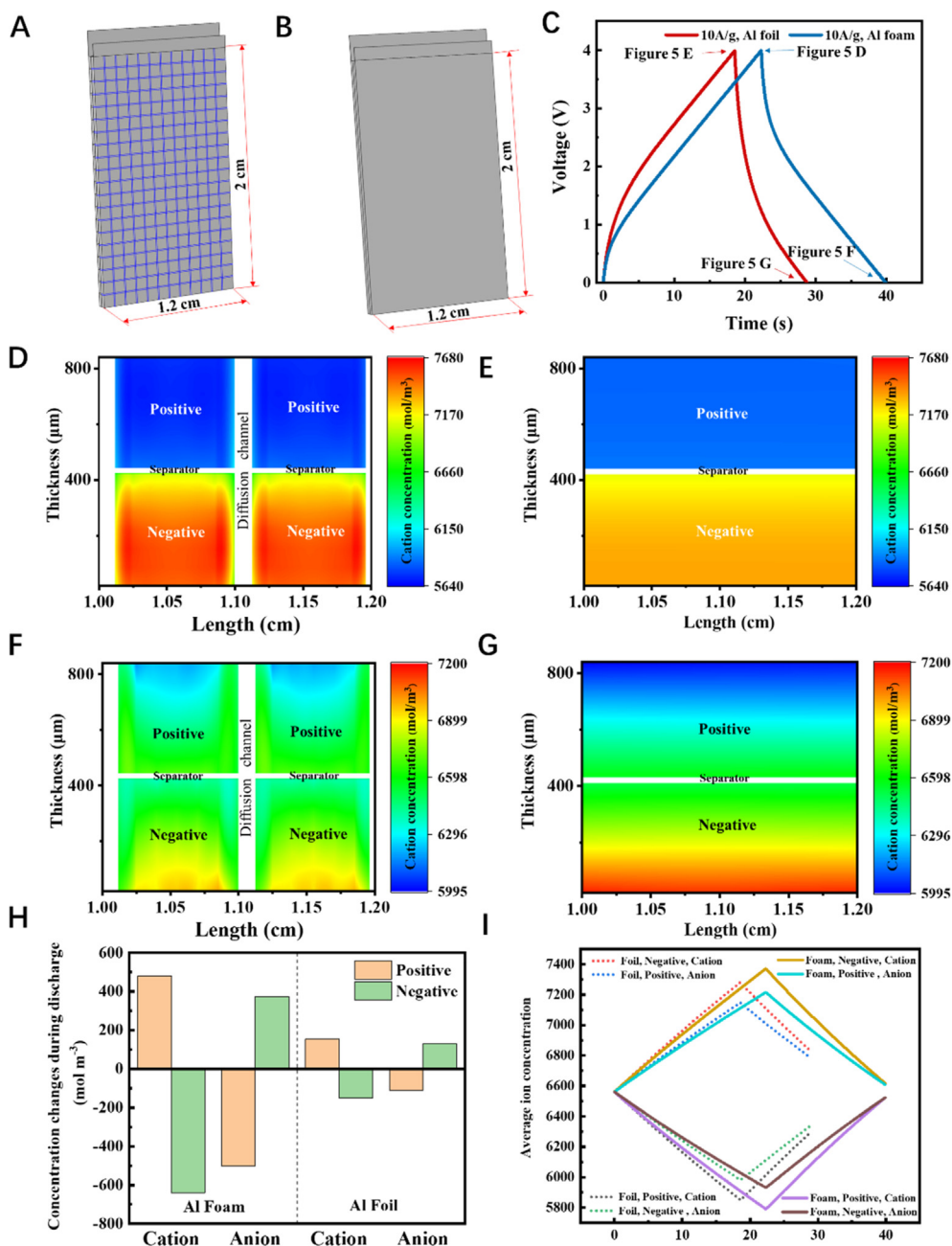


Fig. 5. Simulation results of Al foam and Al foil as current collector SC devices.

(A) 3D model of graphene-Al foam devices. (B) 3D model of graphene-Al foil devices. (C) Simulated voltage curve during the constant current charge and discharge process. (D) Cation distribution in the electrode for graphene-Al foam electrode, charged at 4 V. (E) Cation distribution in the electrode for graphene-Al foil electrode, charged at 4 V. (F) Cation distribution in the electrode for graphene-Al foam electrode, discharged at 0 V. (G) Cation distribution in the electrode for graphene-Al foil electrode, discharged at 0 V. Note that Fig. 5D, E, F, and G is the enlarged region (red box) of Figs. S10A–D (SI-9), respectively. Fig. S10 (SI-9) shows the cation concentration distribution in the full range of side view. The anion distribution of Al foil and Al foam model during the charge and discharge process are plotted in Fig. S11 (SI-9). (H) The changes of maximum concentration in electrode during the discharge process from 4 V to 0 V. (I) The changes of the average ion concentration of the Al foam and Al foil model during charge and discharge process. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

with less diffusion channel. In contrast, the interconnected Al wires, like various cages, broke the continuous layer of graphene, providing sufficient diffusion channel for ions (Fig. 2F). 3D grids were built considering ion diffusion inside 3D foam occurred in multi-direction, driven by the interconnected Al network (Fig. 5A and B).

In general, ions existed on the surface of electrode, pores of electrodes and the bulk phase of electrolyte. Driven by charge, the oriented adsorption of ions on the interface of electrode/electrolyte was kinetically the accessible pores dependent. The thick graphene-Al foil electrode charged quickly approached from 0 V to 4 V at 18.6 s and quickly discharged to 0 V at 28.8 s (Fig. 5C). The type of curve was not symmetric, due to the insufficient discharge. In comparison, it took 22.3 s for graphene inside Al foam charged from 0 V to 4 V and discharged from 4 V to 0 V in 40 s. The discharged time is far longer than that of graphene-Al foil electrode, indicating the rapid diffusion of ions in large amount inside graphene-Al foam electrode contributed to high energy.

Quantitatively, as pure EMIMBF₄ was used as the electrolyte, the ion concentration was around 6516 mol m⁻³, which was too excessive for the 100 F pouch for a very slow charge state. Thus, the ion polarization if any was electrode structure and pore structure-dependent. For two electrodes in the simulation, the largest concentration of ions was not near to the current collector side as charged to 4 V. It was due to the offset effect of random diffusion of ions and the oriented migration of ions driven by charge. Ion concentrations tended to become uniform by the first effect, while tended to exhibit the largest gradient by the second effect. The maximal concentration of ions occurred at the interface of the active electrode with separator. In this region, there still existed a strong diffusion of ions from the direction of the separator to this interface and the diffusion channel. On the other hand, the large diffusion inside the electrode (we define a large Bruggeman coefficient as 3) represented the bottleneck, resulting in the accumulation of ions at the outer edge of active materials.

Fig. 5H showed the changes in the maximum concentration of cation and anion between 4 V and 0 V during the discharge process. Detailed data could be found in Table S4 (SI-9). In Al foam model, during discharge, the concentration of cation in positive electrode was increased by 480 mol m⁻³ while decreased in the negative electrode by 640 mol m⁻³. In contrast, the number of Al foil was only increased by 155 mol m⁻³ in the positive and decreased by 150 mol m⁻³ in the negative, which showed the diffusion channel of Al foam model helped the ion transport more freely under the electric field. It was the same in anion. Considering the ion was not distributed uniformly in the electrode, the mean concentration was also evaluated during the charge and discharge process (Fig. 5I). Both models showed the same concentration at 0 s when the initial voltage was 0 V. During charge, the Al foil model showed a slightly higher average concentration than Al foam model. However, because of the higher ion polarization, the foil model reached 4 V quickly than Al foam model while the concentration of Al foam still increasing, therefore, the Al foam model showed higher average concentration than Al foil at 4 V. During discharge process, the average concentration of Al foam decreased slowly than Al foil model. At the end of discharge (0 V), all the cation and anion concentration were back to the initial state. Most positions of ions became green, returning from the interface to the bulk phase (Fig. 5D and F; Fig. S10A and S10C, SI-9). In comparison, the Al foil model cannot back to its initial state, there were still ions in large amounts without returning to bulk phase (Fig. 5E and G; Fig. S10B and S10D, SI-9). It indicated clearly that the electrochemical polarization of ions in the thick electrode was alleviated significantly by the diffusion channels provided with the Al network. Similar, a study on the thick electrode in battery confirmed the importance of the design of sufficient ion pathways for the better power performance [26]. It explained well why the thick electrode (450 μm) assisted with Al foam exhibited a power density comparable to that of the 120 μm thick electrode-based commercial SC, where active materials were pasted on Al foil [27]. Note that, changing the numbers of diffusion channel, which represented the number of Al wires dividing graphene into different aggregates,

directly influence the accessibility of ions to graphene in the simulation. It, consequently, resulted in the different capacitance and different polarization degrees of ions. Further investigation is needed to build a model much close to the real structure of Al foam assisted graphene electrode.

4. Conclusion

In theory, our fabrication technology with Al foam would be feasible for any small-sized materials. In addition, a similar structure of 3D Ni foam was adopted to host electrode materials in the fabrication of Ni-MH battery (nickel metal hydride battery) by a continuous assembly line with roll-to-roll mode for two decades. The roll-to-roll technology would be also feasible to our 3D Al foam assisted graphene electrode. Such kind of SC would be very attractive for the application in cases with strict limitations of space, e.g. in vehicles and compatible devices.

In summary, we successfully fabricated the 3D Al foam assisted graphene electrode with a mass loading of 10–15 mg cm⁻². Sketch of Al network exerted space confinement effect on graphene and provided sufficient diffusion channels, allowing the smooth volume expansion of the graphene electrode with high mass density. The device based on the graphene-3D Al foam electrodes with 450 μm thick and ionic liquid electrolyte EMIMBF₄, operated at 4 V, exhibited very low contact resistance owing to the uniform structure and tight interaction of Al wires with graphene. The multi-point connection of 3D Al foam with graphene minimized the polarization of electron transfer and ion diffusion, allowing the pouch exhibiting high energy density, high power density and excellent cycling stability simultaneously.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

CRediT authorship contribution statement

Zhoufei Yang: Investigation, Software, Writing - original draft, Visualization. **Jiarui Tian:** Investigation, Writing - original draft. **Zhenzhen Ye:** Investigation. **Ying Jin:** Investigation, Resources. **Chaojie Cui:** Investigation, Writing - original draft, Writing - review & editing. **Qing Xie:** Investigation. **Jin Wang:** Investigation. **Gang Zhang:** Resources. **Zhuoya Dong:** Investigation. **Yonghua Miao:** Resources. **Xiang Yu:** Investigation. **Weizhong Qian:** Conceptualization, Writing - original draft, Writing - review & editing. **Fei Wei:** Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work is financially supported by NSFC program of 21975142 and National key basic research program of 2016YFA0200102.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ensm.2020.07.020>.

References

- [1] Y. Zhu, S. Murali, M.D. Stoller, K.J. Ganesh, W. Cai, P.J. Ferreira, A. Pirkle, R.M. Wallace, K.A. Cychosz, M. Thommes, Carbon-based supercapacitors produced by activation of graphene, *Science* 332 (80) (2011) 1537–1541.

- [2] W. Ren, H.-M. Cheng, The global growth of graphene, *Nat. Nanotechnol.* 9 (2014) 726–730.
- [3] X. Yang, C. Cheng, Y. Wang, L. Qiu, D. Li, Liquid-mediated dense integration of graphene materials for compact capacitive energy storage, *Science* 341 (80) (2013) 534–537.
- [4] R. Raccichini, A. Varzi, S. Passerini, B. Scrosati, The role of graphene for electrochemical energy storage, *Nat. Mater.* 14 (2015) 271–279.
- [5] D.A.C. Brownson, D.K. Kampouris, C.E. Banks, Graphene electrochemistry: fundamental concepts through to prominent applications, *Chem. Soc. Rev.* 41 (2012) 6944–6976.
- [6] J. Yan, S. Li, B. Lan, Y. Wu, P.S. Lee, Rational design of nanostructured electrode materials toward multifunctional supercapacitors, *Adv. Funct. Mater.* 30 (2020), 1902564.
- [7] C. Cui, W. Qian, Y. Yu, C. Kong, B. Yu, L. Xiang, F. Wei, Highly electroconductive mesoporous graphene nanofibers and their capacitance performance at 4 V, *J. Am. Chem. Soc.* 136 (2014) 2256–2259.
- [8] B.G. Choi, M. Yang, W.H. Hong, J.W. Choi, Y.S. Huh, 3D macroporous graphene frameworks for supercapacitors with high energy and power densities, *ACS Nano* 6 (2012) 4020–4028.
- [9] J. Tian, C. Cui, Q. Xie, W. Qian, C. Xue, Y. Miao, Y. Jin, G. Zhang, B. Guo, EMIMBF 4-GBL binary electrolyte working at –70° C and 3.7 V for a high performance graphene-based capacitor, *J. Mater. Chem. A* 6 (2018) 3593–3601.
- [10] Z. Yang, J. Tian, Z. Yin, C. Cui, W. Qian, F. Wei, Carbon nanotube-and graphene-based nanomaterials and applications in high-voltage supercapacitor: a review, *Carbon N. Y.* 141 (2019) 467–480.
- [11] P.J. Hall, M. Mirzaei, S.I. Fletcher, F.B. Sillars, A.J.R. Rennie, G.O. Shitta-Bey, G. Wilson, A. Cruden, R. Carter, Energy storage in electrochemical capacitors: designing functional materials to improve performance, *Energy Environ. Sci.* 3 (2010) 1238–1251.
- [12] C. Schütter, S. Pohlmann, A. Balducci, Industrial requirements of materials for electrical double layer capacitors: impact on current and future applications, *Adv. Energy Mater.* 9 (2019), 1900334.
- [13] B. Yao, S. Chandrasekaran, J. Zhang, W. Xiao, F. Qian, C. Zhu, E.B. Duoss, C.M. Spadaccini, M.A. Worsley, Y. Li, Efficient 3D printed pseudocapacitive electrodes with ultrahigh MnO₂ loading, *Joule* 3 (2019) 459–470.
- [14] W.Z. Qian, J.P. Xue, Z.F. Yang, W.R. You, Y. Jin, S.W. Gu, Porous Al Bulks, the Preparation System and the Method, 2019.
- [15] R. De Levie, Electrochemical response of porous and rough electrodes, *Adv. Electrochem. Electrochem. Eng.* 6 (1967) 329–397.
- [16] C.A. Hall, A. Ignjatovic, Y. Jiang, P.A. Burr, A. Lennon, Time domain modelling of concurrent insertion and capacitive storage using Laplace domain representations of impedance, *J. Electroanal. Chem.* 850 (2019), 113379.
- [17] M. Al Sakka, H. Gualous, N. Omar, J. Van Mierlo, Batteries and Supercapacitors for Electric Vehicles, InTech, 2012.
- [18] Y. Yu, C. Cui, W. Qian, F. Wei, Full capacitance potential of SWCNT electrode in ionic liquids at 4 V, *J. Mater. Chem. A* 2 (2014) 19897–19902.
- [19] L.S. Mtron, LS Ultracapacitor, 2019. <http://www.lsmtron.com/pdf/Ultracapacitor.pdf>.
- [20] Maxwell, Cells Datasheet Documents, 2019. https://www.maxwell.com/images/documents/2_7_1_100_150F_ds_3001957_datasheet.pdf.
- [21] Nippon, Nippon 2019 Catalog, 2019. <https://www.chemi-con.co.jp/e/catalog/pdf/dl-e/dl-all-e1009g-2019.pdf>.
- [22] Panasonic, Electric Double Layer Capacitors (Wound Type), 2017. <https://industrial.panasonic.com/ww/products/capacitors/edlc/edlc-wound-type>.
- [23] Yunasko, Yunasko Products Datasheets, 2019. <https://yunasko.com/en/support>.
- [24] D.J. Finn, M. Lotya, G. Cunningham, R.J. Smith, D. McCloskey, J.F. Donegan, J.N. Coleman, Inkjet deposition of liquid-exfoliated graphene and MoS₂ nanosheets for printed device applications, *J. Mater. Chem. C* 2 (2014) 925–932.
- [25] S. Zhang, X. Lu, Q. Zhou, X. Li, X. Zhang, S. Li, Ionic Liquids: Physicochemical Properties, Book, Elsevier, 2009.
- [26] Y. Kuang, C. Chen, D. Kirsch, L. Hu, Thick electrode batteries: principles, opportunities, and challenges, *Adv. Energy Mater.* 9 (2019), 1901457.
- [27] A. Zhamu, B.Z. Jang, Method of Producing Supercapacitor Electrodes and Cells Having High Active Mass Loading, 2017, 231231.